PROGRAM 16 – FEED FORWARD NEURAL NETWORK

AIM

To write a Python program to implement a Feed Forward Neural Network using input, hidden and output layers with the sigmoid activation function.

ALGORITHM

1. Start the program.
2. Import the NumPy library.
3. Define the input values.
4. Initialize weights and biases for the hidden layer.
5. Calculate the weighted sum of inputs.
6. Apply the sigmoid activation function.
7. Pass the hidden layer output to the output layer.
8. Calculate the weighted sum at the output layer.
9. Apply the sigmoid activation function again.
10. Display the final output.

CODE

```
import numpy as np
def sigmoid(x):
   return 1/(1+np.exp(-x))
inputs=np.array([[1,0]])
weights_hidden=np.array([[0.5,-0.4],[0.3,0.8]])
bias_hidden=np.array([[0.1,0.2]])
weights_output=np.array([[0.7],[-0.6]])
bias_output=np.array([[0.1]])
hidden_input=np.dot(inputs,weights_hidden)+bias_hidden
hidden_output=sigmoid(hidden_input)
final_input=np.dot(hidden_output,weights_output)+bias_output
final_output=sigmoid(final_input)
print("Input Layer:",inputs)
print("Hidden Layer Output:",np.round(hidden_output,4))
print("Final Output:",np.round(final_output,4))
```

INPUT

```
Input Layer: [[1 0]]
```

OUTPUT

```
Input Layer: [[1 0]]
Hidden Layer Output: [[0.6457 0.4502]]
Final Output: [[0.57]]
```

RESULT

Thus, the Feed Forward Neural Network was successfully implemented using NumPy with input, hidden and output layers and the sigmoid activation function.